Talent Retention Strategy: Securing African Expansion

Predictive attrition analysis and strategic workforce stabilization for O-zone Telecom.

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Ambition: Growth vs. Stability

Global Expansion

O-zone Telecom is entering a critical expansion phase in Africa. Success depends on mobilizing and retaining our high-potential talent.

The Retention Crisis

Current high turnover rates threaten operational readiness. Management is split on root causes: Gender Disadvantage vs. Working Conditions.

Methodology: Scientific Integrity

Data Audit

Analyzed 1,518 employee profiles. Validated data integrity with zero missing values and zero duplicates. Overall attrition rate: 16.34%.

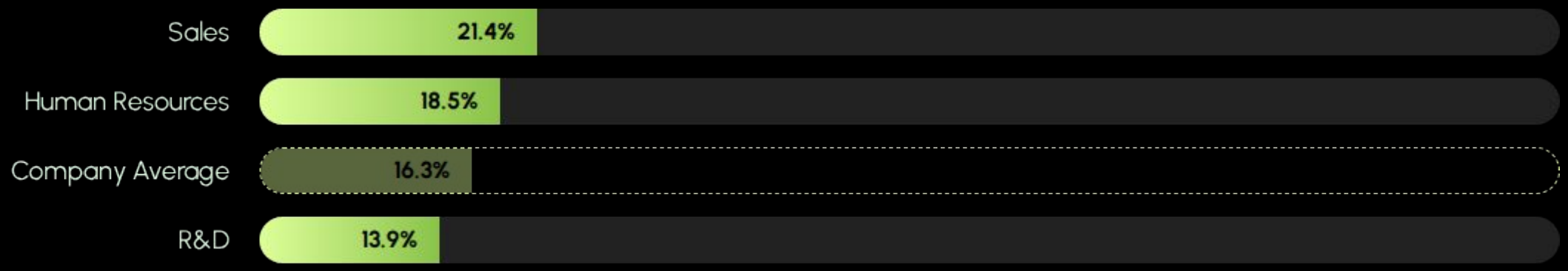
PCA & Clustering

Dimension reduction (PCA) followed by K-Means clustering (K=6) to segment the workforce into distinct risk profiles.

Predictive AI

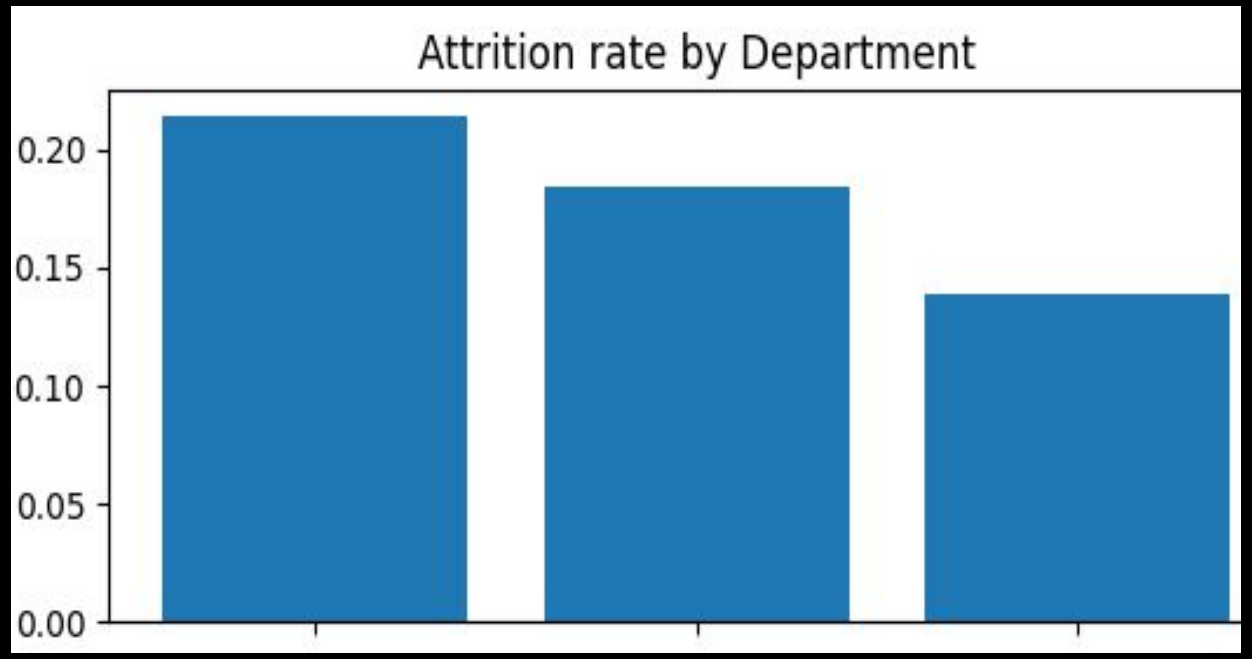
Random Forest model (ROC AUC 0.81) to rank features and predict individual turnover probability with high precision.

Concentration: Sales Risk Exposure



Insight: The Sales department is our primary risk zone (+5 pts vs R&D). Attrition is concentrated in roles critical for the upcoming African market penetration.

Key Driver: Structural Workload



Sales Human Resources R&D

Systemic Diagnostic

Overtime is the primary structural driver of churn. The data refutes gender-only disadvantage as the root cause; rather, workload stress is the main catalyst for talent loss.

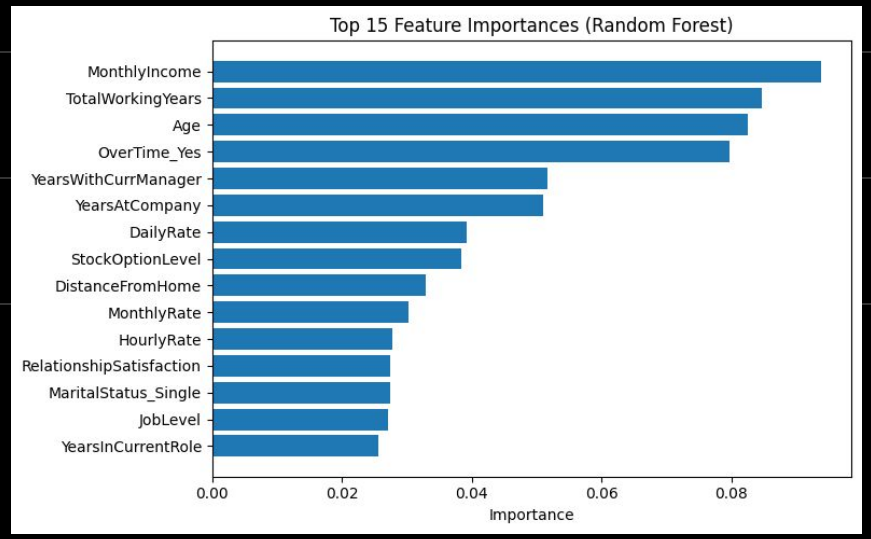
30.1%

Attrition with Overtime

Compared to 10.9% without Overtime

AI Findings: Top 15 Attrition Drivers

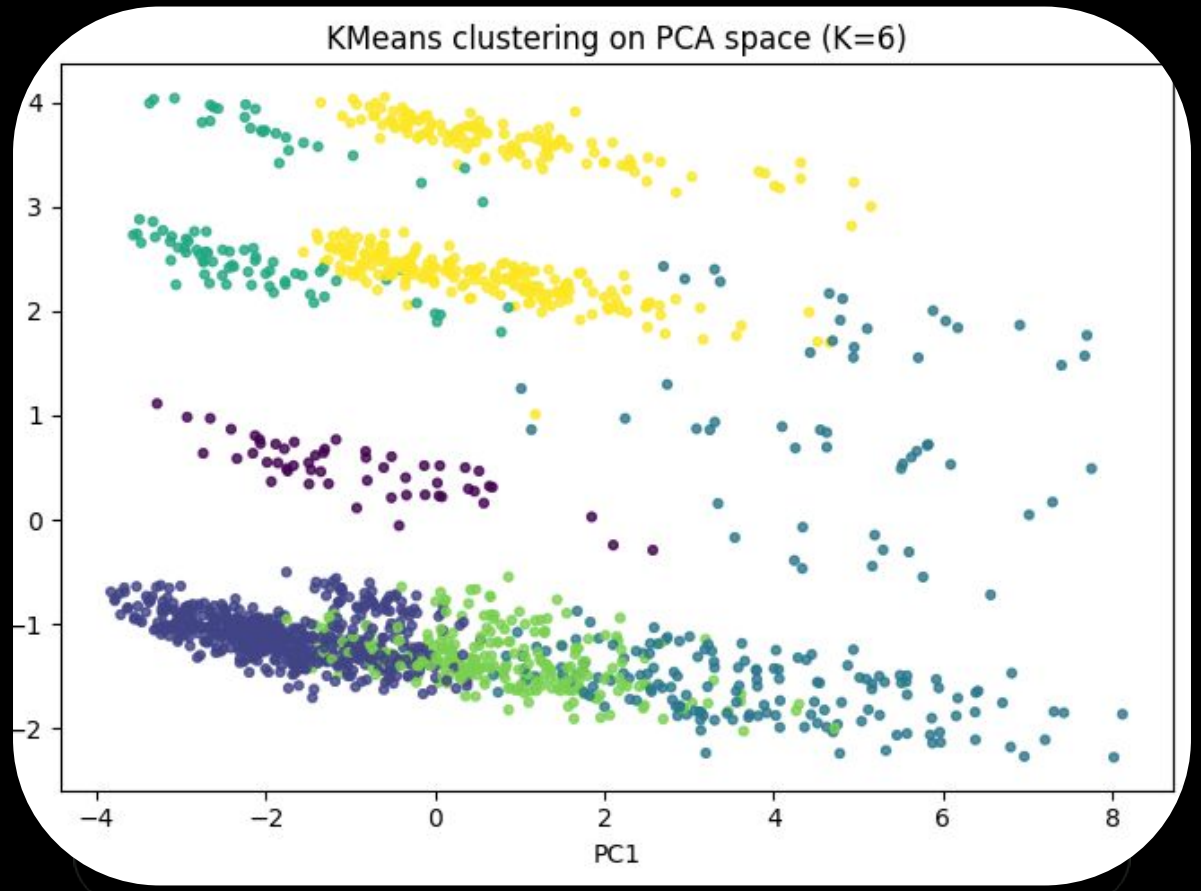
Rank	Feature	Weight
1	Monthly Income	0.093
2	Total Working Years	0.084
3	Age	0.082
4	Overtime Exposure	0.079



Validation

Random Forest analysis highlights that **Compensation** and **Career Maturity** are more predictive than any other factor.
Leavers earn an average of **$2,000 less** per month than stayers.

Segmentation: The Bleeding Segment



Cluster 3 : "At-Risk Juniors"

42% Attrition Rate

- **Income:** Lowest across all segments ($2.6k)
- **Tenure:** Very short (avg. 3 years)
- **Profile:** Early-career, underpaid talent.

This cluster is our structural churn engine, requiring immediate intervention.

Action Plan: 3 Strategic Priorities

1. Wage Realignment

Target Cluster 3 (juniors) for immediate salary review. High cost of turnover justifies a premium retention investment in entry-level roles.

2. Overtime Regulation

Implement strict overtime caps in Sales. Rebalance territory assignments to reduce workload pressure and operational fatigue.

3. Leadership Stability

Incentivize manager continuity for Cluster 0 (mid-level). "Years with Current Manager" is a top stability predictor.

Conclusion: Data to Growth

By shifting from reactive turnover management to **proactive workforce stabilization**, O-zone Telecom will:

- Reduce junior churn by up to **30%**.
- Secure expertise for the **African roll-out**.
- Convert HR insights into **measurable ROI**.

> "Securing our talent today guarantees our global footprint tomorrow."

Questions?

Thank you for your attention.

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